

hw6

chl

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1

Definition 1. $\xi : (\Omega, \mathcal{F}) \rightarrow (S, \mathcal{B}(S))$. S is a metric space. $\mathcal{B}(S)$ is the Borel σ -field on S . $\xi^{-1}(B) \triangleq \{\omega : \xi(\omega) \in B\}$, $\forall B \in \mathcal{B}(S)$.

Prove that the inverse $\xi^{-1}(\cdot)$ has the followings:

1. $\xi^{-1}(S) = \Omega$;
2. $\xi^{-1}(B - C) = \xi^{-1}(B) - \xi^{-1}(C)$, $\forall B \supset C$, $B, C \in \mathcal{B}(S)$;
3. $B \subset C \Rightarrow \xi^{-1}(B) \subset \xi^{-1}(C)$;
4. $\xi^{-1}(A^c) = (\xi^{-1}(A))^c$;
5. $\xi^{-1}\left(\bigcup_{\alpha} B_{\alpha}\right) = \bigcup_{\alpha} \xi^{-1}(B_{\alpha})$;
6. $\xi^{-1}\left(\bigcap_{\alpha} B_{\alpha}\right) = \bigcap_{\alpha} \xi^{-1}(B_{\alpha})$;
7. If $\mathcal{G}$ is a σ -field on S , then $\xi^{-1}(\mathcal{G})$ is a σ -field on Ω ;
8. $\xi^{-1}(\sigma(\mathcal{C})) = \sigma(\xi^{-1}(\mathcal{C}))$, $\mathcal{C}$ is a collection of subsets of S .

Proof. 1. $\forall \omega \in \Omega, \xi(\omega) \in S$, so $\xi^{-1}(S) \supset \Omega$. Together with the trivial fact $\xi^{-1}(S) \subset \Omega$, we have $\xi^{-1}(S) = \Omega$.

2. $\xi^{-1}(B - C) = \{\omega : \xi(\omega) \in B - C\} = \{\omega : \xi(\omega) \in B, \xi(\omega) \notin C\} = \{\omega : \xi(\omega) \in B\} - \{\omega : \xi(\omega) \in C\} = \xi^{-1}(B) - \xi^{-1}(C)$.

3. $B \subset C \Rightarrow \xi^{-1}(B) = \{\omega : \xi(\omega) \in B\} \subset \{\omega : \xi(\omega) \in C\} = \xi^{-1}(C)$.

4. $\xi^{-1}(A^c) = \{\omega : \xi(\omega) \in A^c\} = \{\omega : \xi(\omega) \notin A\} = \{\omega : \xi(\omega) \in A\}^c = (\xi^{-1}(A))^c.$
5. $\xi^{-1}\left(\bigcup_{\alpha} B_{\alpha}\right) = \{\omega : \xi(\omega) \in \bigcup_{\alpha} B_{\alpha}\} = \{\omega : \exists \alpha, \xi(\omega) \in B_{\alpha}\} = \bigcup_{\alpha} \{\omega : \xi(\omega) \in B_{\alpha}\} = \bigcup_{\alpha} \xi^{-1}(B_{\alpha}).$
6. $\xi^{-1}\left(\bigcap_{\alpha} B_{\alpha}\right) = \{\omega : \xi(\omega) \in \bigcap_{\alpha} B_{\alpha}\} = \{\omega : \forall \alpha, \xi(\omega) \in B_{\alpha}\} = \bigcap_{\alpha} \{\omega : \xi(\omega) \in B_{\alpha}\} = \bigcap_{\alpha} \xi^{-1}(B_{\alpha}).$
7. (a) $\xi^{-1}(S) = \Omega \in \xi^{-1}(\mathcal{G});$
 (b) If $A \in \xi^{-1}(\mathcal{G})$, then $\xi^{-1}(A^c) = (\xi^{-1}(A))^c \in \xi^{-1}(\mathcal{G});$
 (c) If $A_n \in \xi^{-1}(\mathcal{G})$ for all n , then $\xi^{-1}\left(\bigcup_n A_n\right) = \bigcup_n \xi^{-1}(A_n) \in \xi^{-1}(\mathcal{G}).$
8.
$$\xi^{-1}(\sigma(\mathcal{C})) = \xi^{-1}\left(\bigcap_{\mathcal{G} \supset \mathcal{C}, \mathcal{G} \text{ is } \sigma\text{-field}} \mathcal{G}\right) = \bigcap_{\mathcal{G} \supset \mathcal{C}, \mathcal{G} \text{ is } \sigma\text{-field}} \xi^{-1}(\mathcal{G}) = \sigma(\xi^{-1}(\mathcal{C}))$$

□

2

Theorem 1. Assume that $\xi : (\Omega, \mathcal{F}) \rightarrow (S, \mathcal{B}(S))$ is a measurable and $\mathcal{B}(S) = \sigma(\mathcal{C})$. Then

$$\xi^{-1}(\mathcal{B}(S)) \subset \mathcal{F} \Leftrightarrow \xi^{-1}(\mathcal{C}) \subset \mathcal{F}.$$

i.e., ξ measurable $\Leftrightarrow \xi^{-1}(\mathcal{C}) \subset \mathcal{F}$.

Proof. $\Rightarrow$: trivial.

$\Leftarrow$: We prove

$$\mathcal{G} \triangleq \{B : X^{-1}(B) \in \mathfrak{F}\} \supset \mathcal{C} \quad (\text{condition})$$

is a σ -field.

1. $X^{-1}(S) = \Omega \in \mathfrak{F} \Rightarrow S \in \mathcal{G}.$
2. $B \in \mathcal{G} \Rightarrow X^{-1}(B) \in \mathfrak{F} \Rightarrow X^{-1}(B^c) = (X^{-1}(B))^c \in \mathfrak{F} \Rightarrow B^c \in \mathcal{G}.$

3. $B_n \in \mathcal{G}, B_n \uparrow B$, then $X^{-1}(B_n) \uparrow X^{-1}(B)$, then $X^{-1}(B) = \bigcup_{n=1}^{\infty} X^{-1}(B_n)$. Since $\mathfrak{F}$ is a σ -field, $X^{-1}(B) \in \mathfrak{F}$, then $B \in \mathcal{G}$.

□

Remark 1. Because $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C}_k)$, $k = 1, 2, \dots, 8$, where

$$\mathcal{C}_1 = \{(-\infty, x) : x \in \mathbb{R}\}, \mathcal{C}_2 = \{(-\infty, x] : x \in \mathbb{R}\},$$

$$\mathcal{C}_3 = \{[a, b) : a, b \in \mathbb{R}\}, \mathcal{C}_4 = \{[a, b] : a, b \in \mathbb{R}\},$$

$$\mathcal{C}_5 = \{(a, b] : a, b \in \mathbb{R}\}, \mathcal{C}_6 = \{(a, b) : a, b \in \mathbb{R}\},$$

$$\mathcal{C}_7 = \{[x, +\infty) : x \in \mathbb{R}\}, \mathcal{C}_8 = \{(x, +\infty) : x \in \mathbb{R}\}.$$

若已知：检验 $\xi : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is r.v. 有 8 种等价的定义

- (i) $\{\omega : \xi(\omega) < x\} \in \mathcal{F} \Leftrightarrow$ (ii) $\{\omega : \xi(\omega) \leq x\} \in \mathcal{F} \Leftrightarrow$ (iii) $\{\omega : \xi(\omega) \in [a, b)\} \in \mathcal{F} \Leftrightarrow$
 (iv) $\{\omega : \xi(\omega) \in [a, b]\} \in \mathcal{F}$
 $\Leftrightarrow$ (v) $\{\omega : \xi(\omega) \in (a, b]\} \in \mathcal{F} \Leftrightarrow$ (vi) $\{\omega : \xi(\omega) \in (a, b)\} \in \mathcal{F} \Leftrightarrow$ (vii) $\{\omega : \xi(\omega) \geq x\} \in \mathcal{F}$
 $\Leftrightarrow$ (viii) $\{\omega : \xi(\omega) > x\} \in \mathcal{F}.$

Proof. We show that $\mathcal{B}(\mathcal{R}) = \sigma(\mathcal{C}_1)$. Trivially, $\sigma(\mathcal{C}_1) \subset \mathcal{B}(\mathcal{R})$. For the other direction, we only need to show that any open set in $\mathcal{R}$ can be generated by $\mathcal{C}_1$.

$\forall$ open set $G \subset \mathcal{R}$, $\forall x \in G$, $\exists$ an open interval $(a_x, b_x) \subset G$ s.t. $x \in (a_x, b_x)$. Then we can write $G = \bigcup_{x \in G} (a_x, b_x)$. By the countable open subcovering theorem, $\exists$ a countable subset $G_0 \subset G$ s.t. $G = \bigcup_{x \in G_0} (a_x, b_x)$.

We next show that any open interval can be generated by $\mathcal{C}_1$. For any $a, b \in \mathcal{R}$, $a < b$, we have $(a, b) = (-\infty, b) \cap (-\infty, a]^c$, and $(-\infty, a] = \bigcap_{n=1}^{\infty} (-\infty, a + \frac{1}{n})$. So $(a, b) = \bigcup_{n=1}^{\infty} (-\infty, b) \cap (-\infty, a + \frac{1}{n})^c \in \sigma(\mathcal{C}_1)$. Therefore, $\mathcal{B}(\mathcal{R}) \subset \sigma(\mathcal{C}_1)$, which implies $\mathcal{B}(\mathcal{R}) = \sigma(\mathcal{C}_1)$.

For other cases, just note that

$$(-\infty, a) = \bigcup_{n=1}^{\infty} (-\infty, a - \frac{1}{n}] \quad (-\infty, a) = \bigcup_{n=1}^{\infty} (a - n, a)$$

□